

Computer Science & IT

Discrete Mathematics



Mathematical Logic

Lecture No. 06



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Recap of Previous Lecture



Topic

Quantifiers



Topic

Scope of the quantifier



Topic

Negation of a statement formula



Topics to be Covered



Topic

Sentences with multiple quantifiers



Topic

Relationship diagram



Sentence with multiple quantifiers $\Rightarrow$

P : "likes" {Domain = set of all students in a class}

$P(x, y) = x \text{ likes } y \equiv y \text{ is liked by } x$

- ① $\forall x \forall y \{P(x, y)\}$: All x likes all y
Every student likes Every student
- ② $\forall y \forall x \{P(x, y)\}$: All y liked by all x
Every student liked by Every student

$$\forall x \forall y \{P(x, y)\} \equiv \forall y \forall x \{P(x, y)\}$$

Sentence with multiple quantifiers $\Rightarrow$

P : "likes" {Domain = set of all students in a class}

$P(x, y) = x \text{ likes } y \equiv y \text{ is liked by } x$

③ $\exists x \forall y \{P(x, y)\}$: Someone likes everyone
Some x likes all y {At least one student likes every student}

④ $\forall y \exists x \{P(x, y)\}$: All y liked by some x
Everybody is liked by somebody {Every student is liked by at least one student}

$$\exists x \forall y \{P(x, y)\} \Rightarrow \forall y \exists x \{P(x, y)\}$$
$$\nLeftarrow$$

→ Let our domain is set of real numbers

$$\rightarrow \exists x \forall y \{x \cdot y = y\} \implies \forall y \exists x \{x \cdot y = y\}$$

it is true w.r.t. $x=1$

w.r.t. ' $x=1$ ' R.H.S. will also be true

$$\rightarrow \forall y \exists x \{x + y = 0\} \not\implies \exists x \forall y \{x + y = 0\}$$

let $y=5$, then $x=-5$

if $y=6$, then $x=-6$

for Every ' y ' there exist at least one ' x ' s.t. $x+y=0$

∴ LHS is true

there exist no ' x ' s.t.

if it is added with any real number then Result will be '0'

but → R.H.S. is false

Sentence with multiple quantifiers $\Rightarrow$

P : "likes" {Domain = set of all students in a class}

$P(x, y) = x$ likes $y \equiv y$ is liked by x

⑤ $\exists y \forall x \{P(x, y)\}$: At least one 'y' is liked by all x
at least one student is liked by all students

⑥ $\forall x \exists y \{P(x, y)\}$: All 'x' likes at least one y
Everyone likes some one

$$\exists y \forall x \{P(x, y)\} \Rightarrow \forall x \exists y \{P(x, y)\}$$

$\nLeftarrow$

Sentence with multiple quantifiers $\Rightarrow$

P : "likes" {Domain = set of all students in a class}

$P(x, y) = x \text{ likes } y \equiv y \text{ is liked by } x$

⑦ $\exists x \exists y \{P(x, y)\}$: At least one student likes at least one student
• Somebody likes somebody

⑧ $\exists y \exists x \{P(x, y)\}$: At least one student liked by at least one student
• Somebody is liked by somebody.

$$\exists x \exists y \{P(x, y)\} \equiv \exists y \exists x \{P(x, y)\}$$

$$\textcircled{1} \quad \forall x \forall y \{ P(x, y) \} \equiv \forall y \forall x \{ P(x, y) \}$$

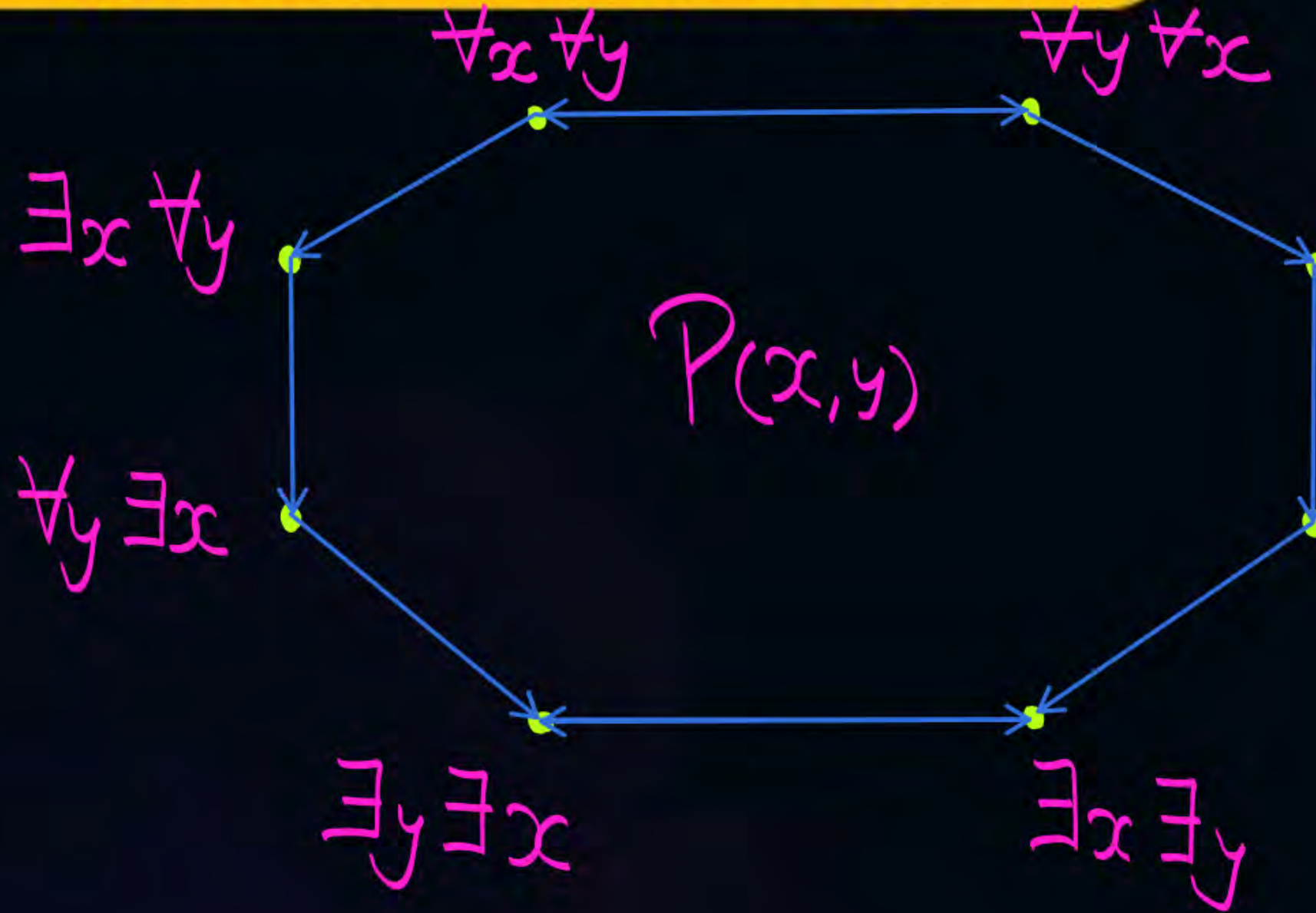
$$\textcircled{2} \quad \exists x \forall y \{ P(x, y) \} \not\equiv \forall y \exists x \{ P(x, y) \}$$

$$\textcircled{3} \quad \exists y \forall x \{ P(x, y) \} \not\equiv \forall x \exists y \{ P(x, y) \}$$

$$\textcircled{4} \quad \exists x \exists y \{ P(x, y) \} \equiv \exists y \exists x \{ P(x, y) \}$$



Topic : Relationship diagram



it represent
Equivalence



it represent
the implication

$\exists y \forall x$

$\forall x \exists y$

#Q. Which of the following is/are true?

- ☒ **A** $\forall x \exists y P(x,y) \Rightarrow \exists y \forall x P(x,y)$
- ☒ **B** $\forall x \forall y P(x,y) \equiv \exists y \forall x P(x,y)$
 $\Rightarrow$ ✓
 $\Leftarrow$ ✗ $\therefore \equiv$ is invalid
- ☒ **C** $\exists x \forall y P(x,y) \Rightarrow \forall y \exists x P(x,y)$
- ☒ **D** $\exists x \exists y P(x,y) \Rightarrow \forall y \exists x P(x,y)$

[MCQ]

$$\forall x (\exists y R(x, y)) \equiv \forall x \exists y (R(x, y)) \equiv \forall x \exists y R(x, y)$$



#Q. Consider the first order logic sentence $F: \forall x (\exists y R(x, y))$. Assuming non-empty logical domains, which of the sentences below are implied by F ?

✓ I. $\exists y (\exists x R(x, y))$

✗ II. $\exists y (\forall x R(x, y))$

✗ III. $\forall y (\exists x R(x, y))$

✓ IV. $\sim \exists x (\forall y \sim R(x, y))$

A IV only

✓ **B** I and IV only

C II only

D II and III only

i.e. whenever F is true, what else is guaranteed to be true

$$\begin{aligned} & \downarrow \text{III} \\ & \forall x (\sim (\forall y \sim R(x, y))) \\ & \forall x \{ \exists y (\sim (\sim R(x, y))) \} \\ & \forall x \exists y (R(x, y)) \equiv F \end{aligned}$$

$P \rightarrow P$
is always true

H.W. Consider following two predicate formula

$$F_1: \quad \forall x \{ P(x) \leftrightarrow Q(x) \} \longrightarrow \{ \forall x \{ P(x) \} \leftrightarrow \forall x \{ Q(x) \} \}$$

$$F_2: \quad \{ \forall x \{ P(x) \} \leftrightarrow \forall x \{ Q(x) \} \} \longrightarrow \forall x \{ P(x) \leftrightarrow Q(x) \}$$

Which of the following is true

- (a) Only F_1 is valid
- (b) Only F_2 is valid
- (c) Both F_1 and F_2 are valid
- (d) Neither F_1 nor F_2 is valid



2 mins Summary



✓ **Topic** Sentences with multiple quantifiers

✓ **Topic** Relationship diagram

THANK - YOU